Measuring the severity of stress scenarios in nats: what entropy offers stress-test designers

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Abstract

Authorities that design stress scenarios, such as the adverse scenario of the EU-wide stress test coordinated by the European Banking Authority (EBA), must answer two questions: is the scenario severe enough, and is it still plausible? This paper measures both on a single scale, entropy, expressed in nats. A nat is a quantity of information. For a pure shift of the mean, a scenario of η nats is a shock of $\sqrt{2\eta}$ standard deviations: one nat buys 1.4 standard deviations. We apply this scale to 33 years of returns on twenty US stocks, to Kenneth French's 48 industry portfolios, and to a euro-area panel covering the five segments of the ECB's Composite Indicator of Systemic Stress (CISS). Three results stand out. **(1)** A simple robustness test, which reruns every computation on data whose time order has been shuffled, removes most of the crisis signals that entropy seems to show: one-sided jumps, fattening tails and the link between tails and correlations are estimation artefacts. The most robust fact, common to every panel, is that markets couple under stress: the number of independent sources of risk falls. Crises are also more often dominated by this coupling, and the index travels further and persists longer than estimation mechanics can explain. **(2)** Measured in nats, classical scenarios that shock volatilities, correlations and levels separately pay 15 to 33 % too much for their tail loss, and the split of the shock between volatility and correlation can be calibrated on history instead of assumed. **(3)** The coupling measure decomposes exactly across market segments. In the euro area it flags recessions as well as the CISS and shows which segments stress ties together, but it does not forecast growth.

1 Introduction

Two facts about financial prices sit uneasily together. First, the Gaussian model is the natural answer to a question of ignorance: fix a mean and a variance, know nothing else, and the maximum-entropy distribution is Gaussian. Second, returns are not Gaussian: they have fat tails and negative skewness, and their dependence tightens precisely when it matters.

The usual reaction treats the second fact as a defect of the first model and patches it: add jumps, add a stochastic volatility factor, add a copula. This paper goes the other way. If the Gaussian is what maximum entropy delivers, any departure from it measures how far the market is from maximal ignorance: it is a measure of information. The question is then not how to patch the Gaussian, but how to read the size, dynamics and direction of the departure.

The paper is built around an asymmetry in this departure. Information arrives in bursts and is absorbed gradually. A central-bank surprise, a default or a war removes uncertainty at once; the market then spends weeks working out the implications, and uncertainty rebuilds. If entropy is the right currency of uncertainty, its path should be a sawtooth: sharp drops, slow recoveries. Half of this shape is a theorem: receiving information cannot, on average, increase uncertainty. The other half, the slow recovery, is a hypothesis we test.

Who this paper is for. It is written first for institutions that design stress scenarios for others: the European Systemic Risk Board (ESRB) and the ECB, which write the adverse scenario of the EBA stress test, national central banks, the IMF in its Financial Sector Assessment Programs, and insurance supervisors. These institutions must produce scenarios that are "severe but plausible" (BCBS, 2018). In practice, severity is measured in losses and plausibility is left to expert judgement. A common unit for both is missing. Entropy provides one: it measures in nats the distance between a scenario and the current state of the market, for any asset class, and it reduces to standard deviations in simple cases. Judging plausibility by a statistical distance is not new (Breuer et al., 2009; Breuer & Csiszár, 2013; Glasserman & Xu, 2014). What we add is a data-based calibration of that distance, a test that separates real signals from artefacts, and an application to the euro area.

Four contributions.

1. *A measurement framework* (section 3). The uncertainty of a return vector splits exactly into three channels: scale (volatilities), dependence (correlations) and shape (skewness and fat tails). Removing scale gives a unit-free index, the structural index $\mathcal{J}$, which measures the market's distance from maximal ignorance.
2. *A robustness test* (sections 4 and 5). An entropy measure estimated on real data jumps up and decays slowly even when no information arrives: this is a mechanical effect of estimation. We therefore judge every result against *placebo nulls*, that is, against the same computations run on resampled data. The most robust result, present on every panel, is that markets couple under stress.
3. *A severity unit for scenarios* (section 6). A scenario of severity η is the worst distribution within η nats of the current one. The radius η is calibrated on how fast the market itself revises its distribution. The same unit lets one audit an existing scenario and choose how to split the shock between volatility and correlation.
4. *A monitoring indicator* (section 7). The coupling measure decomposes exactly across market segments. Applied to the five segments of the ECB's CISS, it shows which segments stress ties together, which directly informs the correlation structure of a scenario.

A warning. The structural index is computed from estimated moments (variances, correlations, skewness, kurtosis). On fat-tailed data, estimation noise pushes such an index *up* abruptly and lets it fall back *slowly*. That is exactly the sawtooth the theory predicts. The result is uncomfortable and, we believe, the most useful thing in the paper: many signals that look overwhelming do not survive the null. Anyone who builds a crisis indicator from estimated moments should run this null before trusting their own chart.

Outline. Section 2 defines the concepts used. Section 3 presents the framework and the eight hypotheses tested. Section 4 describes the data and the placebo test, section 5 the results. Sections 6 and 7 present the two applications: measuring scenario severity and the entropic CISS. Section 8 summarises recommendations and limits. Technical justifications are in the appendix; full proofs are in a separate theoretical supplement.

2 The concepts used

This section defines, in the order they are needed, every concept the rest of the paper uses. No calculation is needed to follow what comes next.

Entropy. The entropy $h(p)$ of a distribution p measures its uncertainty: the more spread out the distribution, the higher its entropy. For a continuous variable, $h(p) = -\int p \log p$. Its level depends on units (expressing returns in per cent rather than decimals shifts it by a constant): only its *changes* and the *divergences* below have an absolute meaning.

Nat. The unit of information obtained with the natural logarithm. 1 nat $\approx$ 1.44 bits. Every quantity in the paper is in nats.

Maximum entropy. Among all distributions with a given mean and covariance, the Gaussian has the highest entropy: it is the distribution that assumes least beyond these two moments. Any departure from the Gaussian is therefore additional information.

Kullback–Leibler (KL) divergence. $D_{KL}(p \,\|\, q) = \int p \log(p/q) \geq 0$ measures in nats the gap between a distribution p and a reference q. It is zero if and only if $p = q$ and does not depend on units. It is not a distance (it is not symmetric), hence the word "divergence". In practice: if two Gaussians share the covariance Σ and have means μ_0 and μ_1, $D_{KL}(p_1 \,\|\, p_0) = \frac{1}{2}(\mu_1 - \mu_0)^\top \Sigma^{-1}(\mu_1 - \mu_0)$, half the squared Mahalanobis distance. A shock of k standard deviations therefore costs $k^2/2$ nats.

Mutual information. $I(X;Y)$ is the KL divergence between the joint law of X and Y and the product of their separate laws. It measures what X and Y share and is zero when they are independent. For two Gaussian variables with correlation ρ, $I = -\frac{1}{2}\log(1-\rho^2)$: 0.83 nat for $\rho = 0.9$.

Total correlation (coupling). For N assets with correlation matrix R, $C(R) = -\frac{1}{2}\log\det R \geq 0$ extends mutual information to more than two variables (Watanabe, 1960). It is zero when the assets are independent and rises when they start moving together. We call it *coupling*. An equivalent reading is the *effective number of risk modes*, which equals N for independent assets and tends to 1 when a single common factor explains everything.

The three channels of entropy. The entropy of a return vector splits exactly into three terms (section 3): *scale* (the level of volatilities), *dependence* (coupling, counted negatively) and *shape* (the deficit defined below).

Entropy deficit $\mathcal{D}$. The KL divergence between the true distribution and the Gaussian with the same mean and covariance. It measures what the Gaussian misses: skewness and fat tails.

Structural index $\mathcal{J}$. The sum of shape and dependence, $\mathcal{J} = \mathcal{D} + C$. It is the KL divergence between the true distribution and independent Gaussians with the same volatilities. It is zero if returns are Gaussian and independent, and it does not depend on the level of volatility.

Stress regime. Dates on which 21-day market volatility exceeds the threshold of the top 10 % observed *up to that date*. The threshold never uses future information.

Placebo null. The same computations, rerun on data whose time order has been shuffled. The *i.i.d.* null draws days one at a time: it keeps fat tails and the cross-asset correlation of each day, and destroys all dynamics. It is repeated 200 times. A result *passes* the null if at most 5 % of the 200 repetitions produce a result as strong as the real data.

Information flow. $I_t^{(h)} = D_{KL}(p_t \,\|\, p_{t-h})$: by how many nats the market has revised its own distribution over h days.

Scenario of severity η and return period. The worst distribution within η nats of the current one. It is given the return period of the information flow of the same size: a radius the market covers in one month only once a decade is a "ten-year" scenario.

Tail loss ($\mathbf{ES}_{99}$). The 99 % expected shortfall: the average loss over the worst 1 % of days.

3 The framework: from maximal ignorance to three channels

3.1 Why the Gaussian is the reference

The step from "maximum entropy gives a Gaussian" to "prices are Gaussian" is often asserted and rarely argued, because maximum entropy takes the variance as given without saying where it comes from. Appendix A closes this gap. If time is measured in traded volume rather than in days, volume fixes the variance, and the least committal distribution of the price increment is Gaussian. This is the mixture-of-distributions hypothesis (Clark, 1973; Ané & Geman, 2000), derived rather than assumed. In calendar time, returns depart from the Gaussian for two reasons: the volume clock is random, or information constrains the market. No empirical result in the paper depends on this bridge; it motivates the Gaussian as the zero point.

3.2 Three channels and the structural index

Write the covariance as $\Sigma = DRD$, where D holds the volatilities σ_i and R the correlations. The entropy of the true distribution p splits exactly (appendix E):

$$h(p) \;=\; \tfrac{N}{2}\log(2\pi e) \;+\; \underbrace{\textstyle\sum_i \log \sigma_i}_{\text{scale}} \;+\; \underbrace{\tfrac{1}{2}\log\det R}_{\text{dependence } (=-C)} \;-\; \underbrace{\mathcal{D}}_{\text{shape}}. \tag{1}$$

The first two terms form the entropy of the Gaussian with the same covariance, the ceiling. The deficit $\mathcal{D}$ says how far below it the true distribution lies.

The scale channel is a problem: it rises with volatility. In a crisis, the entropy computed from the covariance matrix alone *increases* (by 5 nats on our main panel), whereas intuition says a crisis is a time when information constrains the market. Using the level of Gaussian entropy as a stress indicator therefore gives the wrong sign. We remove this channel and keep the other two:

$$\mathcal{J} \;=\; \mathcal{D} \;+\; C \;=\; D_{KL}\!\left(p \,\|\, \textstyle\prod_i q_i\right) \;\geq\; 0, \tag{2}$$

where the q_i are Gaussians with the same volatility as each asset. $\mathcal{J}$ is a true divergence (appendix E): it is non-negative, zero if and only if returns are Gaussian and independent, and unit-free. It rises when assets couple (C) or when their distributions depart from the Gaussian ($\mathcal{D}$), and one can always say which share of a move comes from each channel.

3.3 What the framework predicts, and the eight hypotheses

News reduces uncertainty. Formally, receiving a signal Y lowers the *average* entropy of the return X by exactly the mutual information $I(X;Y)$ (Cover & Thomas, 2006). The entropy jump thus has a measurable size in nats: the information content of the news. The framework adds two postulates, detailed in appendix B. The first (*relaxation*) says that, without new information, $\mathcal{J}$ decays slowly back to its normal level. The second (*sign*) says that jumps come with negative skewness, because $\mathcal{J}$ alone cannot tell a crash from a rally. The result is a sawtooth: $\mathcal{J}$ rises in jumps and decays slowly. The framework also predicts that major news, because it hits every asset on the same day, raises coupling and tail thickness together.

Table 1 turns these predictions into eight testable hypotheses.

	Statement
H1	*Regime signature.* Under stress, coupling rises, the shape deficit rises, hence $\mathcal{J}$ rises.
H2	*Compensation.* When volatilities rise, dependence absorbs part of the entropy increase (slope $\beta > 0$).
H3	*One-sided jumps.* Large moves in $\mathcal{J}$ are rises, small ones are falls: rising episodes are more numerous and larger than falling ones.
H4	*Relaxation.* After an episode, $\mathcal{J}$ decays with a half-life longer than the estimator's memory.
H5	*Sign.* Jumps and rising episodes of $\mathcal{J}$ coincide with negative market skewness.
H6	*Channel coupling.* Dependence and tails move independently in calm markets and together under stress.
H7	*Forecasting.* Today's entropy improves, out of sample, the short-term risk forecast beyond volatility.
H8	*Crisis composition.* In crisis episodes, dependence is more often the dominant channel than in data without dynamics.

Table 1: The eight hypotheses tested. H3, H4, H5 and H8 concern jumps: they are tested day by day and by *episode*, an episode grouping jumps less than 40 days apart.

4 Data and method

4.1 Data

- *Main panel:* twenty individual S&P 500 stocks, daily, 1990–2022 (8,312 days), covering the Asian crisis, LTCM, the dot-com bubble, 2008, the sovereign debt crisis, COVID-19 and 2022. Individual stocks rather than portfolios, because sector aggregation mechanically creates dependence.
- *Industry panel:* Kenneth French's 48 daily industry portfolios, 1990–2026 (about 9,000 days).
- *Control panel:* the DAX, SMI, CAC and FTSE indices, 1991–1998 (Bollerslev & Ghysels, 1996), too small to carry a result on its own.
- *Euro-area panel:* 28 daily series spread across the five CISS segments, 2005–2026 (section 7).

4.2 Estimation

Correlations and volatilities are estimated on a 252-day rolling window with the Ledoit & Wolf (2004) correction, or with exponential weighting (60-day half-life) for everything related to jumps. The shape deficit is estimated asset by asset with the Hyvärinen (1998) approximation, accurate to a few per cent on fat-tailed laws such as daily returns. A jump is flagged when the daily change in $\mathcal{J}$ exceeds four times a jump-robust local scale (Barndorff-Nielsen & Shephard, 2004). Regime comparisons use a block bootstrap with 252-day blocks; forecasts are evaluated out of sample with the Diebold & Mariano (1995) test. Appendix D gives the details.

4.3 Why a placebo null is indispensable

The index $\mathcal{J}$ is a convex, non-negative function of estimated moments. This creates a trap. When an extreme day enters the estimation window, the index jumps; it then decays slowly as that day loses weight. This movement mimics exactly the sawtooth the theory predicts, without any information arriving.

A simulation shows it. Feed independent Gaussian returns with constant covariance, a world with no dynamics at all, through the same pipeline. A single injected stress day raises $\mathcal{J}$ by 2.1 nats, and the index decays with a half-life of 35 days, purely through the estimator's memory. With no event at all, the detector finds 37 positive jumps and no negative ones.

A result is therefore evidence only if it exceeds what data without dynamics produce. Hence the i.i.d. null defined in section 2. It applies exactly the same pipeline, including the definition of the stress regime, and answers the question: beyond the fact that turbulent periods contain big days, does the measure detect any dynamics? A result that passes this null is established.

Why not a block null. One could also draw blocks of 21 consecutive days. Such a null keeps the clustering of turbulent days and all dynamics shorter than a month, and destroys only longer-run dynamics. Yet a market crisis plays out precisely at that scale: COVID raised the index by 16 nats in three weeks, and a typical crisis consists of a few weeks of consecutive turbulent days. A null that keeps these weeks keeps the very phenomenon we want to measure; counting it as "noise" would require the measure to detect something other than the crises themselves. We therefore do not use it as a criterion.

5 Results: what survives the null

Table 2 summarises the verdicts. Detailed figures are in appendix D (tables 11 and 12).

	Raw result (S&P 500)	S&P 500 verdict	FF49 industries verdict
H1 coupling	rise under stress +0.99 nat	**passes**	**passes**
H1 index $\mathcal{J}$	+1.93 nat	fails	null stronger
H1 shape $\mathcal{D}$	+0.27 nat	null stronger	null stronger
H2 compensation	$\beta = 0.20$ under stress	fails	**passes**
H3 jumps, daily	158 rises, 9 falls	null stronger	fails
H3 jumps, by episode	65 rises, 7 falls	null stronger	not tested
H3 largest episode	15.8 nats (COVID)	**passes**	not tested
H4 half-life, daily	121 days	**passes**	**passes**
H4 half-life from peak	18 days (74 for the 5 largest)	fails	not tested
H4 range of $\mathcal{J}$	18.95 nats	**passes**	**passes**
H5 sign, daily	correlation −0.24	fails ($p = 0.10$)	fails
H5 sign, by episode	skewness falls in 62 %	**passes**	not tested
H6 channel coupling	0.04 calm, 0.63 stress	null stronger	null stronger
H7 forecasting, 5–10 days	3.5 % gain ($p = 0.03$ at 5 days)	not robust to multiple testing	**passes** (5 % gain)
H7 forecasting, 21–63 days	zero or negative gain	rejected	rejected
H8 composition	dependence dominant in 46 %	**passes**	**passes**

Table 2: Verdicts against the i.i.d. null (200 repetitions). "Null stronger": the resampled data produce a larger effect than the real data. For H7, the test is out of sample and covers the main and control panels.

5.1 The central fact: under stress, markets couple

Under stress, the coupling of the twenty stocks rises by 0.99 nat. The i.i.d. null produces only 0.21 nat, and none of the 200 repetitions reaches the observed value. In concrete terms, the effective number of independent sources of risk falls from 13.1 to 11.4 out of 20. The same result holds on the 48 French industries: 3.65 nats against 0.56, again with no repetition reaching it. This is the

paper's strongest result, six standard deviations above the null on stocks and twelve on industries. Its presence on both individual stocks and industries rules out an aggregation artefact.

Range and persistence of the index. Real markets also move the index further and for longer than any reshuffling of the same returns. The *range* of $\mathcal{J}$ reaches 18.95 nats on the main panel, against 10.05 under the null, and 54.5 against 36.9 on industries; no repetition reaches it on either panel. The *half-life* of $\mathcal{J}$ measured day by day is 121 days, against 92 under the null, and 190 against 101 on industries, again beyond every repetition. Once the estimator's mechanical memory is removed, a persistence specific to the data remains: the index takes longer to return to its normal level than a world without dynamics allows. This is what the relaxation postulate predicts. We show below that this persistence comes from the succession of turbulent days, not from slower relaxation after each peak.

A jump is not an event. The detector judges *daily* changes in $\mathcal{J}$. A crisis lasting several weeks therefore produces several jumps: COVID yields nine between 24 February and 16 March 2020, while the index rises by 16 nats, whereas the largest single-day jump is only 3.2 nats. We therefore also count by *episode*, merging jumps less than 40 days apart: the 158 jumps of the main panel form 65 episodes. Measured this way, the largest shock (15.8 nats, COVID) clearly exceeds the null: 8.1 nats on average, and none of the 200 repetitions reaches it. This result matches the one on the range of $\mathcal{J}$ and rests on a single episode. Jumps also cluster more than under the null (2.4 per episode against 2.0, $p = 0.015$): turbulent days follow one another, which is the very signature of a lasting crisis. The five largest episodes (7.9 nats on average against 6.3, $p = 0.025$) also pass the null. The average episode (1.4 nat), however, is smaller than under the null: most real episodes are modest, and only a few are exceptional.

Coupling is not just a volatility effect. Three checks, in table 3, rule out the usual objections.

- *Volatility.* Measured correlations rise mechanically when variances rise (Forbes & Rigobon, 2002). We therefore divide each return by its recent volatility before computing correlations ("devolatilisation"). Coupling falls by about 30 % but still passes the null, by more than five standard deviations, on both panels.
- *Extreme days.* We replace each return by its rank, converted to a Gaussian score, which makes the measure insensitive to extreme values. This "rank" version is the strongest of the three (9.6 and 12.7 standard deviations).
- *Existing measures.* The absorption ratio of Kritzman et al. (2011), which measures the share of variance explained by the first factors, fails the null on the twenty stocks ($p = 0.215$) and passes only on industries. The share of the first eigenvalue passes on both panels, but by a smaller margin than coupling. The likely reason is that $\log\det R$ uses the whole spectrum of the correlation matrix, including the small eigenvalues that collapse when assets couple.

5.2 What does not survive

- *One-sided jumps (H3).* The detector finds 158 upward jumps against 9 downward, and the chart of $\mathcal{J}$ is visibly a sawtooth (figure 2 in the appendix). But the i.i.d. null produces as many: 148 rises and 3 falls on average, with a stronger asymmetry than in the data. The sawtooth is what a fat-tailed law produces once passed through the estimator, whether or not information arrives in bursts. Counting by episode changes nothing: 65 rising episodes

Measure	S&P 500, 20 stocks	FF49, 48 industries
Coupling C, raw	6.2 (< 0.005)	11.9 (< 0.005)
Coupling C, devolatilised	5.1 (< 0.005)	7.9 (< 0.005)
Coupling C, rank	**9.6** (**< 0.005**)	**12.7** (**< 0.005**)
Absorption ratio	0.7 (0.215)	6.9 (< 0.005)
First-eigenvalue share	5.3 (< 0.005)	6.7 (< 0.005)

Table 3: Which dependence measure best separates stress from calm? For each measure, we compute its average rise under stress relative to calm periods and compare it with the same rise under the i.i.d. null. The figure gives by how many standard deviations the observed rise exceeds the null's: the larger it is, the more the measure detects stress beyond chance. In parentheses, the p-value (share of the 200 null repetitions that do at least as well).

against 7 falling, but the null is even more one-sided (31 rises per fall on average, against 9 in the data).

- *Shape under stress (H1, $\mathcal{D}$).* Tails fatten under stress (+0.27 nat), but less than under the null (+1.25). A "stress" window in a shuffled world contains isolated extreme days, which fatten measured tails more.
- *Relaxation after each peak (H4, by episode).* The overall persistence of $\mathcal{J}$ exceeds the null (previous section), but episode by episode, from peak to halfway back, relaxation takes a median of 18 days (74 for the five largest episodes), no longer than under the null (23 days; 68 for the five largest). Persistence therefore comes from the succession of turbulent days, not from slower absorption of each shock.
- *Daily sign (H5).* Jumps in $\mathcal{J}$ do coincide with negative skewness (correlation -0.24), but the null produces the same effect in 10 % of cases, above the 5 % threshold. By episode, however, market skewness falls between the start and the peak in 62 % of episodes, against 50 % under the null ($p = 0.015$): the sign is established at the episode scale, not the daily one.
- *Channel coupling (H6).* Dependence and tails move together under stress (0.63 against 0.04 in calm), but the null couples them even more (0.90). A window containing one big common day shows both more correlation and fatter tails, real or shuffled.
- *Compensation (H2).* The compensation slope ($\beta = 0.20$) does not exceed the null on the twenty stocks ($p = 0.40$); it does on industries (0.23 against 0.08, $p = 0.03$), where coupling weighs more. The result is therefore not common to both panels.

5.3 A weak short-term predictive power (H7)

The question is: does knowing entropy today say anything about risk over the next few days that today's volatility does not already say? We compare a model that uses only volatility with one that adds an entropy channel. Both are estimated on the past only, re-estimated every year, and compared on dates they have not seen, spaced by the forecast horizon so that forecast periods do not overlap (Diebold & Mariano, 1995 test). We repeat the test at four horizons: 5, 10, 21 and 63 trading days (table 4).

The result depends on the horizon. At 5 and 10 days, the unhealed-shock indicator, which measures the share of the last peak of $\mathcal{J}$ not yet absorbed, improves the forecast by 3 to 5 % on both panels, and in the same direction for drawdown and worst loss. The gain shrinks at 21 days,

Horizon (trading days)	5	10	21	63
S&P 500, 20 stocks	−3.5 % (0.03)	−3.5 % (0.09)	−3.3 % (0.23)	+1.0 % (1.00)
FF49, 48 industries	**−5.4 % (0.001)**	−5.2 % (0.009)	−4.2 % (0.09)	+4.2 % (0.16)

Table 4: Change in the mean squared error of the realised-volatility forecast when the unhealed-shock indicator ($\mathcal{J}_t / \max_{252\,\mathrm{d}} \mathcal{J}$) is added to volatility; negative = better forecast. In parentheses, the Diebold–Mariano p-value. In bold: the only case that remains significant after a Bonferroni correction for the 15 tests of one horizon (5 indicators, 3 targets).

where it is no longer significant, and reverses at 63 days: at long horizons, adding entropy worsens the forecast. The signal is therefore real but weak and short-lived, consistent with an index that jumps and then heals within a few weeks. Two caveats limit its reach: only the FF49 case at 5 days survives the multiple-testing correction, and the 21-day horizon had been set before the others. Finally, in sample, all coefficients look highly significant ($p < 10^{-4}$): this is the overlapping-window trap, which would suggest a much stronger predictive power. Entropy measures are above all *diagnostic*, with a modest short-term predictive contribution.

5.4 How crises split their intensity (H8)

The previous hypotheses compare averages over a whole regime. H8 looks at each crisis. We group jumps less than 40 days apart into *episodes* and split the rise in $\mathcal{J}$ in each episode exactly into dependence, skewness and tails. Skewness almost never dominates (1 episode out of 139). The split between dependence and tails varies: dependence dominates 46 % of the 65 episodes of the main panel, against 26 % on average under the null (range 18–33 %, no repetition reaches it); and 79 % of the 61 industry episodes, against 43 %. The largest episode of each panel also gives more weight to dependence than under the null (45 % against 33 %, $p = 0.015$, on stocks).

The result matches the coupling result: when turbulent days follow one another, what distinguishes a real crisis from a random cluster of big days is that assets start moving together. It underpins the calibration of scenario composition (section 6.5). The size of an episode does not predict its composition (rank correlation 0.05).

Takeaway. What distinguishes a real crisis from a series of big days is not the fattening of each asset's tails; it is that assets couple, that crises are more often dominated by this coupling, and that the index goes further and stays high longer than a world without dynamics allows. For a scenario designer, this means the dependence structure of a scenario deserves as much attention as the size of the shocks, and that any crisis indicator built from estimated moments must be checked against a null before use.

6 Application 1: measuring and calibrating scenario severity

This section relies on none of the hypotheses H1–H7: it is a measurement convention plus an optimisation. It speaks directly to the design of market scenarios, such as the financial shocks of the European adverse scenario or the global market shocks of the US tests.

6.1 An entropy-ball scenario

Definition 1 (Scenario of severity η)**.** Let p_0 be the current return distribution and ρ a risk measure (for example the ES_{99} of a portfolio). The scenario of severity η is the most adverse distribution within η nats of p_0:

$$p^\star_\eta \;=\; \arg \max_{p\,:\,D_{KL}(p\,\|\,p_0)\,\le\,\eta} \rho(p). \tag{3}$$

The construction is known (Breuer & Csiszár, 2013; Glasserman & Xu, 2014). For a portfolio and a Gaussian reference it has an explicit solution: the scenario shifts the portfolio mean down and raises its volatility, both drawn from the same budget of η nats. The simplest case gives the exchange rate between information and risk:

$$\text{a purely directional shock of } \eta \text{ nats is a move of } \sqrt{2\eta} \text{ standard deviations.} \tag{4}$$

One nat buys 1.41 standard deviations, two nats buy two, and a three-standard-deviation shock costs 4.5 nats. The severity parameter is therefore not an abstraction: it translates directly into the language of committees.

6.2 Calibrating the radius on the market itself

It remains to choose η. We read it from the market's information flow, that is, from how much the market itself revises its distribution over 21 days, $I_t^{(21)} = D_{KL}(p_t \,\|\, p_{t-21})$, measured on the same quantity as the constraint (the portfolio distribution). Historical quantiles of this flow give a radius for each return period (table 5).

Return period	1 yr	2 yrs	5 yrs	10 yrs	20 yrs	50 yrs†
radius η (nats)	0.017	0.045	0.149	0.408	0.634	1.369
directional shock (std. dev.)	−0.08	−0.13	−0.24	−0.38	−0.47	−0.66
volatility multiplier	1.12	1.20	1.37	1.63	1.80	2.21
one-day ES_{99}	3.9%	4.2%	4.9%	5.9%	6.6%	8.3%
historical: realised volatility increase	1.54	1.77	2.28	3.25	3.57	4.43

Table 5: Severity ladder for an equally weighted portfolio of the twenty stocks, calibrated on the 21-day information flow (383 non-overlapping observations). Current state: annual volatility 20 %, one-day ES_{99} 3.4 %. Last row: quantile, at the same return period, of the ratio of next-21-day volatility to past-252-day volatility. † Extrapolated beyond 33 years of data.

The scenario spends its budget mostly on volatility rather than direction: at the ten-year rung, 63 % more volatility and a 0.38-standard-deviation shift, because ES is more sensitive to scale than to location. The radius depends on the portfolio: a ladder must be calibrated for the portfolio it is applied to.

A serious limit. The last row confronts the ladder with history. A ten-year scenario multiplies volatility by 1.6, whereas the ten-year quantile of the realised volatility increase is 3.2. The ladder is therefore *too mild* by a factor close to two. Either the information flow, measured on smoothed estimators, understates the true revision, or a Gaussian ball is the wrong geometry for such abrupt volatility shocks. We do not settle between the two. The return periods in the table must therefore not be quoted as frequencies. What withstands this limit are the *relative* comparisons of the next two subsections.

6.3 Auditing an existing scenario

The same unit prices a scenario built by expert judgement. Take the type of scenario common in stress exercises: all volatilities multiplied by the same factor, all correlations forced to a high value, and a directional shock of several standard deviations. Its cost in nats, $D_{KL}(p_{\text{scenario}} \,\|\, p_0)$, has a closed form. It can then be compared with the entropy-ball scenario that produces *the same tail loss* (table 6).

Classical scenario	moderate	standard	severe
volatility multiplier	1.5	2.0	3.0
all correlations forced to	0.70	0.90	0.95
directional shock	-2σ	-3σ	-4σ
resulting one-day ES_{99}	9.1%	13.7%	20.3%
cost seen by the portfolio (nats)	2.75	7.24	16.22
cost of the entropic scenario with the same ES_{99}	**1.85**	**5.47**	**13.74**
excess cost of the classical scenario	33 %	24 %	15 %
Cost of each component alone, on the 20 assets (nats)			
directional shock	2.00	4.50	8.00
volatility dilation	4.39	16.14	58.03
forced correlations	2.43	8.10	13.50
full scenario	6.80	9.97	20.19

Table 6: Cost in nats of classical scenarios, compared with the entropic scenario that produces the same tail loss. The "cost seen by the portfolio" is the divergence on the portfolio distribution, the only one comparable to the radius of a ball. For reference, the largest radius observed in 33 years is about 2 nats, and the worst day in the sample costs 9 nats.

Three readings.

- *The classical scenario is too expensive.* For the same tail loss, it costs 15 to 33 % more nats than necessary (6 to 15 % on the French industries). The reason is structural: it also distorts directions the portfolio does not see.
- *The expensive component is uniform volatility dilation.* Shifting the mean by three standard deviations costs 4.5 nats. Doubling every volatility costs 16.1 nats on its own, more than the whole scenario.
- *Volatility and correlation pay for each other.* The three components cost 28.7 nats separately but 10.0 together. Forcing correlations concentrates risk on a single market factor, and dilating that factor is far cheaper than dilating every direction. This is exactly what observed crises do: they load risk onto a few common factors (section 5). A scenario that shocks volatility and correlation as two separate sliders cannot represent this.

For an authority, the practical benefit is the ability to *audit* an existing scenario without replacing it: each scenario receives a cost in nats, comparable across exercises and asset classes, and one can check that it does not buy its severity through a needlessly improbable route.

6.4 Example: the EBA 2025 adverse scenario in nats

We apply the measure to the adverse scenario of the 2025 EU-wide stress test, designed by the ESRB and published by the EBA (EBA, 2025). Its first year, the most severe, lowers EU equities by 50 %, raises long-term rates by 110 basis points in Germany and 168 for the EU, and leaves

euro exchange rates unchanged. Read as a shift in the mean of one-year changes, this shock costs $\frac{1}{2}\Delta^\top\Sigma^{-1}\Delta$ nats, where Σ is the historical covariance of the same variables on the euro-area panel (detailed calculation and table 10 in appendix C).

The scenario costs 10.5 nats, a 4.6-standard-deviation shock on a single factor. It is thus *severe but plausible* in the literal sense: as improbable as 2008 (11.2 nats) or 2022 (11.9) as they actually happened, less than the worst observed year (14.4, March 2008 to March 2009), and far more than 2011 (3.1). It does use dependence: taken one by one, the five indices and two rates would cost 31.9 nats, but national indices move almost as a single factor (average correlation 0.87). Finally, the choice of Σ is a severity choice: with a crisis covariance (2008–2012), the same scenario costs only 8.4 nats. The measure makes this choice explicit.

6.5 Composing a scenario: volatility or correlation?

The entropy ball fixes the severity *of the portfolio* (its stressed mean and volatility). It does not say how each asset's covariance should move to get there. The default choice, used so far, is a uniform dilation of volatilities with correlations unchanged. That is a convention, not a fact about crises. Yet section 5 shows that, under stress, dependence moves on its own.

A simple property allows for this without changing severity. Portfolio variance, $w^\top\Sigma w$, is linear in Σ. If two covariances give the same portfolio variance, so does any mixture of the two (proposition in appendix E). One can therefore mix an "all-volatility" covariance and an "all-correlation" covariance with a weight α, the *dependence share*, without touching severity. One caveat: correlation alone cannot carry any severity. With all correlations equal to one, our portfolio's volatility cannot exceed 1.5 times its current level; beyond that, the rest comes from dilation.

The dependence share is calibrated on the robust results. The *anchor* comes from H1: under stress, dependence accounts for $0.99/(0.99 + 5.97) = 14\,\%$ of the move in Gaussian entropy (scale plus dependence). The *refinement* comes from H8: over the more severe half of episodes, this share reaches 49 %.

Composition of the ten-year scenario ($\eta = 0.41$)	dependence share	total cost (nats)
default: uniform dilation	0.00	6.80
calibrated on H1 (regime average)	0.14	2.88
refined by H8 (severe episodes)	0.49	0.71

Table 7: Total cost in nats, on the twenty assets, of three compositions that produce exactly the same portfolio severity ($\mathrm{ES}_{99} = 5.9\,\%$). On the French industries: 14.7, 8.3 and 5.8 nats for shares of 0, 0.20 and 0.75.

The result is counter-intuitive (table 7): the larger the dependence share, the *less* improbable the scenario, for the same severity. Uniform dilation is therefore not a prudent choice; it is both less realistic and more costly in plausibility than a composition anchored on history. This finding is specific to this panel and should not be read as a general law.

6.6 Reverse stress testing

The same unit also answers the reverse question: what is the least improbable scenario that produces a given loss? On the main panel, the daily loss exceeded one day in a hundred (3.2 %) costs 1.5 nats; the one exceeded one day in a thousand (7.7 %) costs 4.8 nats; the worst day in 33 years (11.5 %) costs 9.0 nats. Beyond a few nats, the non-parametric version of this calculation rests on a handful of days, and the Gaussian version should be preferred.

7 Application 2: an entropic CISS

The paper's one robust result, coupling under stress, becomes here an indicator in the format of the ECB's CISS (Holló, Kremer & Lo Duca, 2012). Like the CISS, it covers the money market, bond, equity, financial intermediary and foreign exchange segments. Unlike the CISS, it says which share of coupling lies within each segment and which lies *between* segments. For a scenario designer, this is precisely the information needed to set correlations between shocks to different markets.

7.1 An exact decomposition by segment

Split the series into segments $B_1, \ldots, B_K$. Total coupling decomposes exactly:

$$C(R) \;=\; \underbrace{\sum_{k=1}^{K} C(R_{kk})}_{\text{within segments}} \;+\; \underbrace{I(B_1; \ldots; B_K)}_{\text{between segments}}, \tag{5}$$

where R_{kk} is the correlation matrix of segment k. The between-segment term is zero exactly when the segments move independently of one another: it is the systemic part. For two segments A and B, the mutual information $I(A;B)$ gives a contagion map in the same unit. Everything is unit-free: equity returns, rate changes and exchange rates can sit in the same matrix. We use the rank version of coupling.

7.2 The euro-area panel

The panel is daily, from January 2005 to October 2026 (5,690 days), with 28 series: the overnight rate (EONIA, then €STR) for the money market; six ECB sovereign yields and spreads for bonds; five national indices for equities; nine banks and three insurers for intermediaries; four euro exchange rates. The CISS uses the Euribor minus overnight spread for the money market; since Euribor is not available on the ECB portal, this segment for now contains only the overnight rate.

7.3 Results

Component	Rise under stress (nats)	Null (mean)	p
Total coupling	+1.61	+0.21	< 0.005
Between segments	**+0.19**	+0.02	$<$ **0.005**
Within: equities / intermediaries	+0.46 / +0.48	+0.05 / +0.14	< 0.005
Within: bonds / FX	+0.34 / +0.14	0.00 / 0.00	< 0.005
Equities – intermediaries	+0.096	+0.011	< 0.005
Bonds – FX	+0.055	+0.004	< 0.005
Bonds – intermediaries	+0.040	+0.003	< 0.005
Money market – other segments	≈ 0	≈ 0	n.s.

Table 8: Euro area: stress-minus-calm rise against the i.i.d. null (200 repetitions). Stress is the top decile of equity-block volatility, with the threshold computed on the past only.

Three readings (table 8 and figure 1).

- *Stress ties segments together*, far beyond the null (the between-segment rise is nine times the null's). Every pair outside the money market rises; the links between equities and intermediaries and with FX are the clearest.

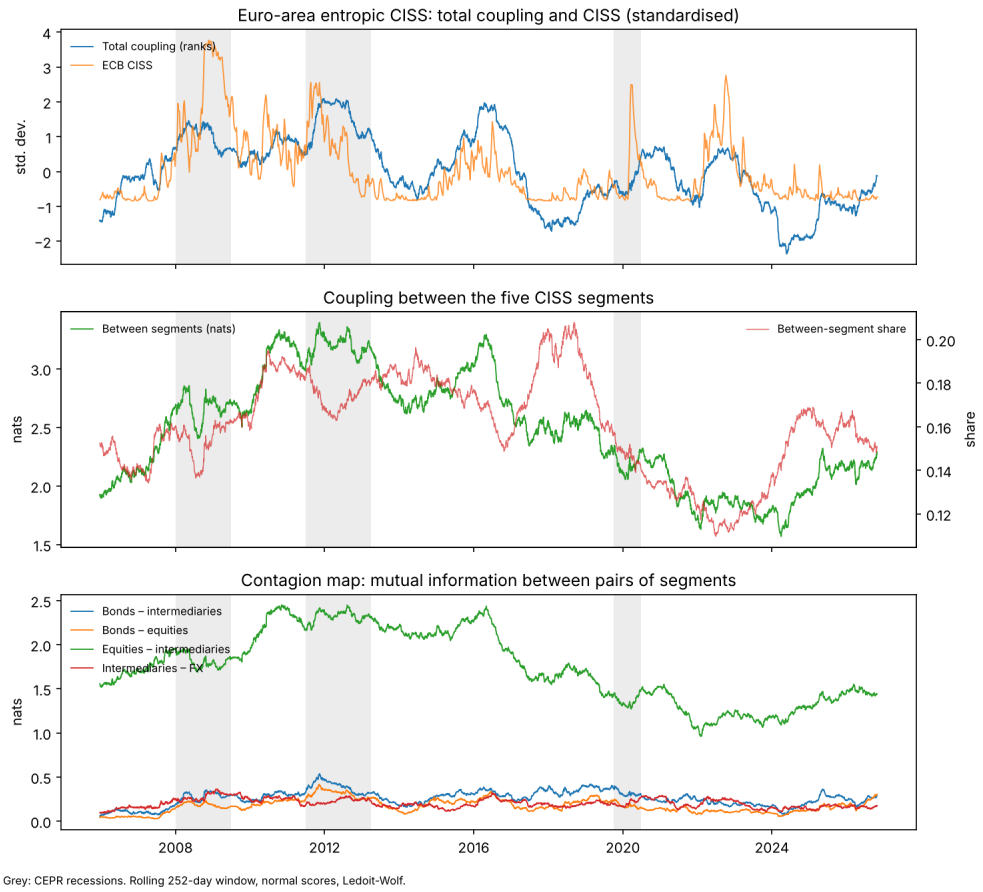


Figure 1: The euro-area entropic CISS. Top: total coupling and the ECB's CISS, standardised. Middle: coupling between the five segments, in nats, and its share of the total. Bottom: mutual information between pairs of segments. Grey areas: CEPR recessions.

- *The sovereign–bank link stands out.* Mutual information between bonds and intermediaries peaks in late 2011, at the height of the sovereign debt crisis (0.46 nat against 0.10 in late 2006). The aggregate CISS cannot show which segments an episode ties together.
- *The money market does not couple* in daily data: the overnight rate follows monetary policy, not market stress.

	Total coupling	Between segments	Volatility	CISS
Correlation with the CISS	0.58	0.35	0.74	1
AUROC, CEPR recessions	**0.85**	0.75	0.80	0.83
Explains the CISS beyond volatility (t)	**6.7**	1.7		
Growth-at-Risk, alone: p	0.04	< 0.001	0.98	< 0.001
Growth-at-Risk, with CISS and past growth: p	0.78	0.81		< 0.001

Table 9: Euro area against official indicators. The AUROC is the probability that a randomly drawn recession date has a higher indicator than a non-recession date (0.5: no power; 1: perfect separation). Growth-at-Risk: regression of the bottom decile of four-quarter GDP growth on the indicator (Adrian, Boyarchenko & Giannone, 2019), 79 quarters.

Against official indicators (table 9), total coupling flags recessions slightly better than the CISS and volatility, and it explains the CISS beyond what volatility explains. It does not, however, forecast growth once the CISS is accounted for. With three recessions and 79 quarters, these comparisons are indicative. The indicator is therefore *coincident*: a complement to the CISS that says which segments are linked, not a better predictor. On US data, against the St. Louis Fed stress index (Kliesen & Smith, 2010), coupling adds to volatility on the twenty stocks ($t \approx 4$ to 5), mainly before 2015, but not on the 48 industries.

Next steps. Add Euribor from its administrator (EMMI) to rebuild the money-market segment as in the CISS; date stress episodes with the ESRB crisis database (Lo Duca et al., 2017) and measure how the contagion map behaves around them; compare the indicator with institution-level systemic-risk measures, SRISK (Brownlees & Engle, 2017) and ΔCoVaR (Adrian & Brunnermeier, 2016).

8 Recommendations and limits

8.1 Recommendations for scenario designers

1. *Price every scenario in nats.* The cost $D_{KL}(p_{\text{scenario}} \,\|\, p_0)$ has a closed form for a Gaussian scenario. It makes scenarios on different markets and exercises comparable, and reads in standard deviations (k standard deviations $= k^2/2$ nats).
2. *Check efficiency before severity.* Compare a scenario's cost with that of the entropic scenario with the same loss: an excess cost of 15 to 33 %, as for the classical scenarios tested here, signals severity bought through a needlessly improbable route.
3. *Do not dilate volatilities uniformly by default.* Shocking part of the severity through dependence is both more faithful to observed crises and less improbable for the same severity.
4. *Set cross-market correlations from a coupling map.* The decomposition (5) says which segments stress ties together, and by how much, in a common unit.
5. *Run a placebo null* before adopting any crisis indicator built from estimated moments.

8.2 Limits

- *Only some predictions survive.* Coupling under stress, crisis composition, and the range and persistence of the index pass the null on both panels. One-sided jumps, fattening tails, the link between tails and correlations, and the daily sign are not established.
- *A single null.* The i.i.d. null destroys all dynamics. It does not say whether the detected dynamics exceed those produced by mere volatility clustering at the monthly scale; we count that as part of the phenomenon the measure should capture (section 4.3).
- *Return periods are not validated.* The severity ladder is about twice too mild against history (table 5). Only the relative comparisons (excess cost, composition) withstand this flaw.
- *Modest predictive power.* The measures are mainly diagnostic: their contribution to forecasting is 3 to 5 % at 5–10 days, nil beyond, and survives multiple testing on one panel only.
- *Shape is estimated asset by asset.* Nonlinear tail dependence is left out of $\mathcal{J}$, which therefore understates the true divergence, especially under stress.

- *Cross-sections are small.* Twenty stocks or 48 industries are not enough to make the reported figures constants; a scenario radius is not comparable across asset universes.
- *The entropic CISS is incomplete.* Its money-market segment awaits Euribor, and its comparisons rest on three recessions.

9 Conclusion

If the Gaussian is the distribution of maximal ignorance, any departure from it measures information, in nats. This idea yields an exact decomposition of a market's uncertainty into three channels, and a unit-free index, $\mathcal{J}$, that measures the distance from maximal ignorance.

The empirical result is first a warning. Every prediction of the framework is visible in the raw data: the index rises under stress, moves in a sawtooth, its jumps are one-sided and come with negative skewness. Rerun on data whose time order has been shuffled, almost all of it disappears, and several effects are even stronger in the shuffled data. Only one thing durably separates a crisis from a series of big days: markets couple.

For institutions that design stress scenarios, the framework offers three tools. A unit, the nat, which measures a scenario's distance from the current state of the market and reads in standard deviations. An audit, which shows that classical scenarios pay 15 to 33 % too much for their tail loss, because they dilate volatilities uniformly instead of coupling markets as crises do. And a map, the entropic CISS, which says which segments stress ties together: a complement to the CISS, not a predictor.

A The bridge from order flow to Gaussian prices

Over an interval, let V be traded volume and Q net signed order flow (buys minus sells). Two assumptions suffice. *Linear impact*: the change in log price is λQ, as in Kyle (1985). *Variance proportional to volume*: given V, the market knows only the mean of Q and a variance $\varsigma^2 V$. Maximum entropy then gives

$$\Delta X \mid V \;\sim\; \mathcal{N}\big(\lambda m,\; \lambda^2 \varsigma^2 V\big)\,. \tag{6}$$

The conditioning variable is volume, not time: this is the mixture-of-distributions hypothesis of Clark (1973) and Tauchen & Pitts (1983), and Ané & Geman (2000) show that returns measured in transaction time are close to Gaussian. Unconditionally, the return is a mixture of Gaussians, hence fat-tailed (Mandelbrot & Taylor, 1967): fat tails alone do not prove that information is present. If increments are moreover independent in volume time, the log price is a time-changed Brownian motion, and the price a geometric Brownian motion. Both assumptions are imperfect (real impact is closer to a square root, and order flow is autocorrelated), which is why all empirical work is done in calendar time.

B Index dynamics: a theorem and two postulates

The theorem. For a return X and a signal Y, $\mathbb{E}_Y[h(X \mid Y)] = h(X) - I(X;Y) \le h(X)$ (Cover & Thomas, 2006). Receiving information lowers entropy *on average* by exactly the mutual information. A particular realisation can raise it (a confusing signal); the jump component is therefore mainly, not exclusively, one-sided.

Postulate 1 (relaxation). Without new information, $\mathcal{J}$ reverts to a long-run level: $d\mathcal{J}_t = -\kappa(\mathcal{J}_t - \bar{\mathcal{J}})\,dt + \sigma_{\mathcal{J}}\,dW_t$. An Ornstein–Uhlenbeck process is the only process that is Markovian, Gaussian and stationary (Doob, 1942). Adding the jumps caused by news gives

$$d\mathcal{J}_t \;=\; -\kappa(\mathcal{J}_t - \bar{\mathcal{J}})\,dt \;+\; \sigma_{\mathcal{J}}\,dW_t \;+\; dZ_t, \tag{7}$$

where Z_t is the sum of positive jumps up to t: the class of Barndorff-Nielsen & Shephard (2001). The half-life of a shock is $\ln 2/\kappa$.

Postulate 2 (sign). $\mathcal{J}$ is an even function of skewness: a distribution with a long left tail (a crash) and its mirror image (a rally) have the same $\mathcal{J}$. Saying anything about direction requires an outside ingredient. We postulate that jumps come with negative skewness, relying on the leverage effect (Black, 1976; Engle & Ng, 1993).

Why dependence and tails should rise together. Major news hits every asset on the same day. On that day, assets move together (coupling rises) and each records an unusual loss (its tails fatten). The two channels describe the same day seen from two places. In calm markets, large moves are asset-specific and couple nothing. This is H6. The null shows, however, that this coincidence is also produced mechanically by estimation.

C Computing the cost in nats of the EBA 2025 adverse scenario

Code: scripts/eba_nats.py; results: paper/tables/eba_nats.csv.

1. The shocks. We take the adverse macro-financial scenario (EBA, 2025) for 2025, as deviations from the baseline (annual averages): EU equities −50 %, the baseline assuming them unchanged (table 4.1.7 of the document); German long rate 3.50 % against 2.40 % and EU long rate 4.80 % against 3.12 % (table 4.1.6); euro exchange rates against the dollar, pound, yen and Swiss franc unchanged (table 4.1.10). Each shock is mapped to the corresponding series of the euro-area panel (section 7), giving an eleven-component vector Δ:

EBA shock	Panel series	Δ_i	σ_i (1 yr)	cost alone (nats)
EU equities −50 %	DAX, CAC 40, FTSE MIB, IBEX 35, AEX	$\ln 0.5 = -0.693$	19–23 %	4.4–6.5
German long rate	10-year AAA yield (ECB)	+1.10 pt	0.65 pt	1.4
EU long rate	10-year all-ratings yield (ECB)	+1.68 pt	0.68 pt	3.0
FX unchanged	EUR/USD, GBP, JPY, CHF	0	7–11 %	0

Equities as log returns, rates as level changes; the maturity of long rates is not specified by the document and is taken at ten years. GDP, unemployment, real estate and inflation, which are not market prices, are not used.

2. The reference. $p_0 = \mathcal{N}(0, \Sigma)$, where Σ is the covariance of one-year changes in the eleven series, estimated on daily changes from 6 September 2004 to 31 December 2024 (the data available to the designer) and multiplied by 252.

Cost in nats	equities and rates	with FX fixed
EBA 2025 adverse scenario, year 1	**10.5**	**11.1**
equities only / rates only	6.8 / 3.1	
sum of variable-by-variable costs	31.9	
Observed years: 2011 / 2008 / 2022	3.1 / 11.2 / 11.9	3.1 / 18.2 / 13.2
worst rolling year (March 2008–March 2009)	14.4	19.5

Table 10: Cost in nats of the EBA 2025 adverse scenario and of observed years, on five equity indices, two ten-year rates and four exchange rates, with the annual covariance estimated on 2004–2024.

3. The formula. The scenario fixes means, not volatilities or correlations. We read it as a pure mean shift, hence (appendix E) $D_{KL}(p_{\text{scenario}} \,\|\, p_0) = \frac{1}{2}\Delta^\top \Sigma^{-1} \Delta$. This is the *minimal* cost consistent with the scenario: among all distributions with mean Δ, the shifted Gaussian is the closest to p_0. Unspecified variables (two- and five-year rates, banks, etc.) are left free; filling them with their conditional expectation is the cheapest choice, and the cost then depends only on the shocked block S: $\frac{1}{2}\Delta_S^\top \Sigma_{SS}^{-1} \Delta_S$. The cost of a single variable is $\frac{1}{2}(\Delta_i/\sigma_i)^2$; for the DAX ($\sigma = 20.3\,\%$), the shock is 3.4 standard deviations and costs 5.8 nats. Read η nats as $\sqrt{2\eta}$ standard deviations.

4. Reading. The sum of variable-by-variable costs is 31.9 nats, the joint cost 10.5: a common shock to highly correlated indices follows the most likely direction in the data. Equities alone (6.8) plus rates alone (3.1) leave 0.5 nat of interaction (falling equities with rising rates go against the historical correlation, +0.13). Fixing FX adds 0.7 nat.

5. Observed years. Same calculation, same Σ, on the realised changes of the eleven series (31 December to 31 December; worst 252-day window ending 17 March 2009). The median year costs 4.1 nats. The scenario combines the equity fall of 2008 (when rates fell) and the rate rise of 2022 (close to 3 points).

6. Variants (equities and rates / with FX fixed). Covariance of rolling annual changes: 12.8 / 18.7; sample to 2026: 10.6 / 11.3; crisis covariance 2008–2012: 8.4 / 9.0. At the end of the three-year horizon (2027: $-42\,\%$, +87 and +112 bp, three-year covariance), the cost falls to 2.1 / 2.2: severity is concentrated in the first year.

7. Limits. Shocks in annual averages; the EU equity fall applied to each index; a Gaussian reference without fat tails, which tends to overstate the cost. The ECB's instantaneous market-risk scenario is not priced (detailed tables not accessible), but the same formula applies. These costs (eleven series, one year) are not directly comparable with table 5 (equity portfolio, 21 days).

D Empirical details

Estimators. Two covariance estimators run in parallel: a 252-day rolling window with the Ledoit & Wolf (2004) correction, and exponential weighting with a 60-day half-life and 10 % shrinkage towards the diagonal. The second prevents an extreme day *leaving* the window from creating a false jump. The shape deficit uses the Hyvärinen (1998) approximation, whose two terms separate a skewness channel and a tail channel; the better known Edgeworth expansion $S^2/12 + K^2/48$

overstates the deficit of a Student t with 4 degrees of freedom by a factor of 69. Jumps are detected at four times a local scale given by bipower variation lagged by one day (Barndorff-Nielsen & Shephard, 2004).

Inference. Two consecutive rolling windows share 251 of their 252 days. Regime comparisons therefore use a circular block bootstrap with 252-day blocks. The relaxation p-value follows the Dickey–Fuller distribution. Forecasts (H7) are re-estimated every year on the past and compared on dates spaced by the forecast horizon, with the Diebold & Mariano (1995) test.

Raw stress-regime results (S&P 500). Under stress (1,076 dates against 6,481), scale rises by 5.97 nats, dependence falls by 0.99, shape rises by 0.27 ($p = 0.16$), and $\mathcal{J}$ rises by 1.94 ($p < 0.001$, but without passing the null). Average correlation goes from 0.26 to 0.32, the effective number of modes from 13.1 to 11.4. Market skewness does not change significantly between regimes: the skewness signature is a property of jumps, not of the regime. The index shares about a fifth of its variance with volatility ($r = 0.46$), and the shape channel is orthogonal to it ($r = 0.08$).

			i.i.d. null			
	Statistic	Obs.	mean	5–95 %	p	Verdict
H1	gap H^{dep}	−0.99	−0.21	−0.42–−0.00	< 0.005	**passes**
H1	gap $\mathcal{J}$	1.93	2.14	1.59–2.79	0.665	fails
H1	gap $\mathcal{D}$	0.27	1.25	0.82–1.75	1.000	wrong side
H2	corr($\Delta H^{vol}, \Delta H^{dep}$)	0.10	0.23	−0.26–0.60	0.295	fails
H2	β (stress)	0.20	0.05	−0.33–0.31	0.395	fails
H3	positive jumps	158.0	148.5	136.9–161.0	0.135	fails
H3	negative jumps	9.0	3.0	1.0–6.0	1.000	wrong side
H3	mean positive jump (nats)	0.62	1.12	0.95–1.29	1.000	wrong side
H3	skewness of $\Delta\mathcal{J}$	12.26	14.80	12.27–17.60	0.950	wrong side
H3	var. share of pos. jumps	0.838	0.899	0.869–0.922	0.990	wrong side
H4	half-life (days)	120.8	92.1	82.4–103.3	< 0.005	**passes**
H4	sd($\mathcal{J}$)	2.03	1.55	1.31–1.81	0.010	**passes**
H4	range($\mathcal{J}$)	18.95	10.05	8.05–12.88	< 0.005	**passes**
H5	gap ΔS_{mkt}	−0.053	−0.011	−0.088–0.070	0.175	fails
H5	corr($\Delta\mathcal{J}, \Delta S_{mkt}$)	−0.243	−0.035	−0.298–0.230	0.100	fails
H6	coupling, stress	0.63	0.90	0.84–0.94	1.000	wrong side
H6	coupling gap	0.590	0.769	0.671–0.871	0.990	wrong side
–	corr($\mathcal{J}$, log vol)	0.461	0.414	0.359–0.466	—	—

Table 11: S&P 500: observed statistics against the i.i.d. null, 200 repetitions. p: share of repetitions at least as extreme in the predicted direction. "Wrong side": the null exceeds the data. Gaps: stress minus calm.

			i.i.d. null			
	Statistic	Obs.	mean	5–95 %	p	Verdict
H1	gap H^{dep}	−3.65	−0.56	−1.03–−0.19	< 0.005	**passes**
H1	gap $\mathcal{J}$	5.93	7.95	6.12–10.11	0.960	wrong side
H1	gap $\mathcal{D}$	0.98	4.81	3.47–6.39	1.000	wrong side
H2	corr($\Delta H^{vol}, \Delta H^{dep}$)	−0.63	−0.36	−0.54–−0.19	< 0.005	**passes**
H2	β (stress)	0.23	0.08	−0.05–0.21	0.030	**passes**
H3	positive jumps	182.0	181.8	165.0–201.1	0.495	fails
H3	negative jumps	3.0	7.8	4.0–13.0	0.025	**passes**
H3	mean positive jump (nats)	1.70	3.47	2.74–4.18	1.000	wrong side
H3	skewness of $\Delta\mathcal{J}$	17.04	18.31	14.84–21.83	0.695	fails
H3	var. share of pos. jumps	0.889	0.943	0.918–0.958	0.980	wrong side
H4	half-life (days)	189.8	101.0	89.7–112.1	< 0.005	**passes**
H4	sd($\mathcal{J}$)	6.37	5.46	4.43–6.48	0.065	fails
H4	range($\mathcal{J}$)	54.47	36.86	28.53–43.95	< 0.005	**passes**
H5	gap ΔS_{mkt}	−0.112	−0.057	−0.140–0.017	0.110	fails
H5	corr($\Delta\mathcal{J}, \Delta S_{mkt}$)	−0.440	−0.197	−0.472–0.075	0.095	fails
H6	coupling, stress	0.76	0.94	0.91–0.96	1.000	wrong side
H6	coupling gap	0.411	0.635	0.510–0.757	1.000	wrong side
–	corr($\mathcal{J}$, log vol)	0.677	0.443	0.388–0.490	—	—

Table 12: FF49 industries, 1990–2026: same conventions as table 11.

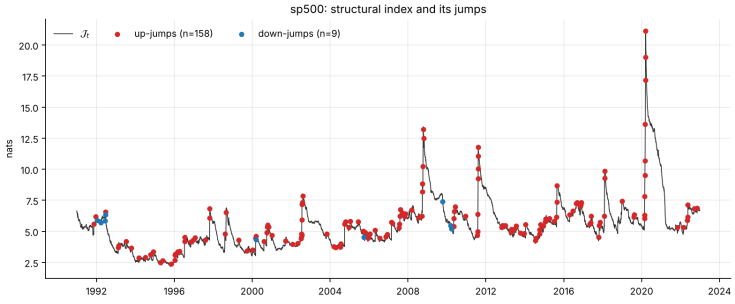


Figure 2: The structural index $\mathcal{J}_t$ on twenty S&P 500 stocks, 1990–2022, with detected jumps (red: rises, blue: falls). The sawtooth is visible, but the i.i.d. null reproduces it.

E Key proofs

Gaussian entropy and equation (1). For $q = \mathcal{N}(\mu, \Sigma)$, $h(q) = \frac{1}{2}\log\big((2\pi e)^N \det\Sigma\big)$. With $\Sigma = DRD$, $\det\Sigma = (\prod_i \sigma_i)^2 \det R$, hence $h(q) = \frac{N}{2}\log(2\pi e) + \sum_i \log\sigma_i + \frac{1}{2}\log\det R$. Since $\det R$ is the product of the eigenvalues of R, which sum to N, the arithmetic–geometric mean inequality gives $\det R \leq 1$: correlation always destroys entropy.

The deficit is a divergence. If q is the Gaussian with the mean and covariance of p, then $\mathbb{E}_p[\log q] = \mathbb{E}_q[\log q]$ because $\log q$ is quadratic. Hence $D_{KL}(p \,\|\, q) = h(q) - h(p) = \mathcal{D} \geq 0$, with equality if and only if p is Gaussian.

The structural index is a divergence. With q_i the Gaussians with the same means and volatilities as each asset, $D_{KL}(p \,\|\, \prod_i q_i) = -h(p) + \sum_i \log\sigma_i + \frac{N}{2}\log(2\pi e)$. Substituting (1) for $h(p)$ leaves $\mathcal{D} - \frac{1}{2}\log\det R = \mathcal{D} + C = \mathcal{J}$.

Mixing compositions. If $w^\top\Sigma_a w = w^\top\Sigma_b w = s^2$, then for every $\alpha \in [0,1]$, $w^\top\big((1-\alpha)\Sigma_a + \alpha\Sigma_b\big)w = s^2$, by linearity.

The price of a directional shock. For two Gaussians with the same covariance, $D_{KL}(p_1 \,\|\, p_0) = \frac{1}{2}\Delta^\top\Sigma^{-1}\Delta$, where Δ is the difference in means. A shock of k standard deviations on a portfolio therefore costs $k^2/2$ nats, and η nats buy $\sqrt{2\eta}$ standard deviations.

Decomposition by segment. Applying the definition of C to each block,

$$C(R) - \textstyle\sum_k C(R_{kk}) = \frac{1}{2}\big(\sum_k \log\det R_{kk} - \log\det R\big),$$

which is the Gaussian mutual information between the blocks, non-negative by Fischer's inequality.

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